

Learning Day 7B Coverage Completion

Classification foundations, Decision Tree theory, KNN practical, and final documentation

Final output: 3 notebooks + 5 documentation artifacts

Customer Segmentation dataset | Decision Tree MVD + missing theory + KNN practical exercise

Why This Work Was Needed

MVD was not enough - uploaded theory and missing KNN topics needed full coverage

Problem

Some uploaded theory was not fully covered earlier.

MVD was only part of the assignment.

Fix

Created separate missing-topic notebooks.

Mapped theory and practical work together.

Outcome

Coverage now includes foundations, Decision Tree theory, pruning, KNN, imputation, and comparison.

Coverage Structure

Three notebooks plus original Decision Tree MVD work

Notebook	Main Coverage	Status
Notebook 1	Classification foundations, dataset, EDA, algorithm families	Complete
Notebook 2	Decision Tree theory, split metrics, pruning, boundaries	Complete
Notebook 3	KNN theory, practical model, tuning, imputation	Complete
Original MVD	Decision Tree practical exercise and SHAP/oblique attempt	Complete

Dataset Overview

Same Customer Segmentation dataset used consistently

Dataset

Train.csv: 8,068 rows, 11 columns

Target: Segmentation

Classes: A, B, C, D

Key EDA

Segment D: youngest avg age

Segment D: 87.04% Low Spending

Segment C: highest Average Spending share

Why it fits

Business-friendly

Multiclass classification

Good for Decision Tree and KNN comparison

Decision Tree Theory Covered

Rules, split quality, pruning, and overfitting control

Area	Covered Concept
Tree structure	Root node, internal node, branch, leaf, decision path
Split quality	Gini, Entropy, Information Gain, Gain Ratio
Search behavior	Hypothesis space, greedy search, no backtracking
Model simplicity	Inductive bias, Occam Razor, MDL
Pruning	Pre-pruning, post-pruning, ccp_alpha, subtree concepts

KNN Theory Covered

Distance-based classification from zero

Core Idea

KNN predicts a customer segment by finding similar old customers.

The nearest customers vote for the predicted class.

Key Concepts

k value

Euclidean distance

Majority voting

Distance-weighted voting

Lazy learning

Practical Needs

Missing values handled

Categorical columns encoded

Numeric columns scaled

Irrelevant ID removed

KNN Preprocessing Pipeline

Clean numeric input for distance calculation

Column Type	Action	Why
Numeric	Median imputation + StandardScaler	Fill missing values and make distance fair
Categorical	Most frequent imputation + OneHotEncoder	Convert text labels to numeric columns
Target	Keep labels A/B/C/D	Prediction output

KNN Model Results

Baseline vs tuned model

Experiment	Train Accuracy	Test Accuracy	Gap	Interpretation
Baseline KNN k=5	0.6264	0.4796	0.1469	Some overfitting
Best KNN k=21 uniform	0.5555	0.5204	0.0350	Best generalization
k=21 distance voting	0.9558	0.5025	0.4534	Strong overfitting

Segment-Level KNN Performance

Best: Segment D | Weakest: Segment B

Segment	Precision	Recall	F1-score	Interpretation
A	0.372	0.442	0.404	Moderate
B	0.348	0.331	0.339	Weakest
C	0.544	0.520	0.532	Good
D	0.654	0.599	0.625	Best

KNN Imputation Mini-Lab

Median imputation vs KNN imputation

Median Imputation

Fills missing numeric values with one middle value.

Simple, stable, easy to explain.

KNN Imputation

Looks at similar rows and creates more customized estimates.

Needs numeric scaled data.

Project Decision

Main model used median imputation for stability. KNNImputer was covered as concept mini-lab.

KNN vs Decision Tree

Two different ways to solve classification

Area	Decision Tree	KNN
Learning style	Eager learner	Lazy learner
Logic	If-then rules	Similar old customers
Scaling needed	No	Yes
Interpretability	High	Medium
Main risk	Overfitting if too deep	Sensitive to scale, noise, dimensions

Final Learning Outcome

Coverage complete; packaging created

What is now covered

Decision Tree MVD
Classification foundations
Decision Tree missing theory
KNN missing theory and practical

Best KNN setting

n_neighbors = 21
weights = uniform
test accuracy = 0.5204
train-test gap = 0.0350

Final message

Decision Tree explains with rules.
KNN explains with similar customers.
Both need different modelling judgement.